

Table 1: How decentralised storage differs from centralised storage

Feature	Centralised storage	Decentralised storage
Control	Managed by one provider (like AWS, Google)	Shared across a network of participants (like IPFS, Filecoin, Arweave)
Data location	Stored in specific locations, often known as data centres	Distributed across multiple nodes globally
Single point of failure	Centralised systems are prone to single points of failure; if the provider experiences downtime, access to data can be lost	Its systems avoid failure by redundantly storing data across multiple nodes
Censorship risk	The risk of censorship is high since providers can remove or block data	Censorship risk is low due to distributed control
Privacy	Depends on the provider's policy	Data is encrypted and often anonymous by design
Cost	Monthly fees; pay for storage and bandwidth	Often pay once or based on actual usage

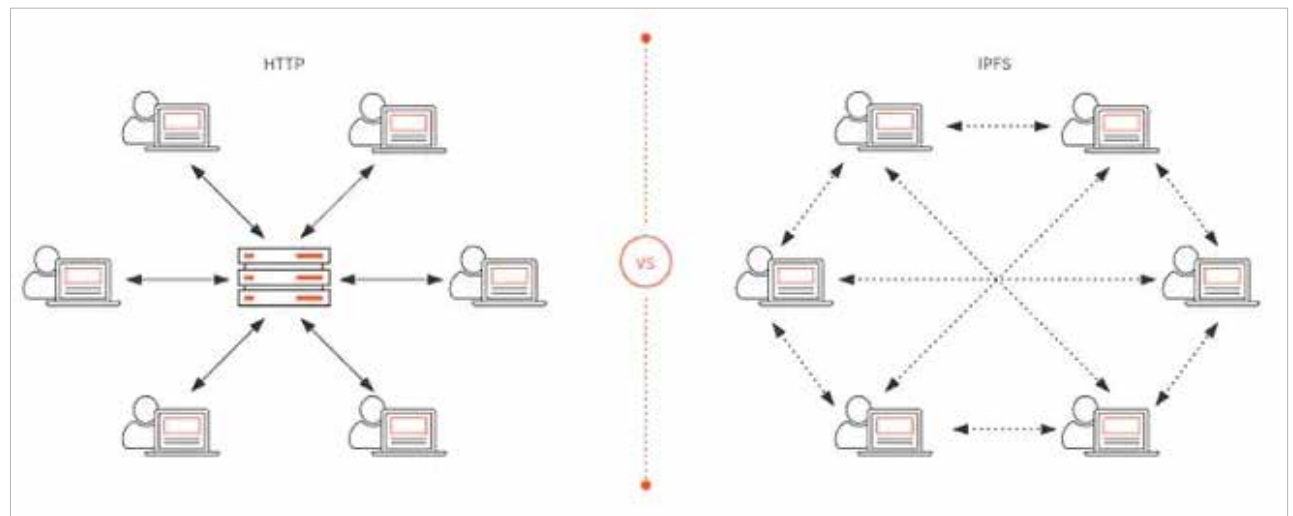


Figure 2: InterPlanetary File System

centralised web (based on HTTP) with a decentralised, content-addressable protocol. By shifting from location-based URLs to content-based identifiers, IPFS makes the web faster, more secure, and resistant to censorship.

How it works: IPFS operates by breaking files into smaller blocks and distributing them across a decentralised network. Here's a step-by-step breakdown.

Step 1. Chunking: First, files are divided into multiple blocks or chunks.

Step 2. Hashing: Then each block is hashed using cryptographic algorithms.

Step 3. Content Identifier (CID):

Next, these hashes generate a unique CID for each piece of data.

Step 4. Retrieval: When you request a file, IPFS retrieves the data from any peer in the network that has the corresponding hash, and not from a centralised server.

Key features

Content-addressable: Files are identified by their content and not by their location, ensuring immutability and verifiability.

Deduplication: Identical blocks of data are stored only once, which significantly reduces redundancy.

Offline-first capability: If you

already have the data or hash stored locally then you can access it without requiring an internet connection.

Limitations: IPFS lacks a built-in economic model to encourage storage. Nodes must voluntarily 'pin' or host files to ensure availability. If a file is not pinned or hosted by any node, it may eventually disappear from the network.

Filecoin

Filecoin, also developed by Protocol Labs, is a decentralised storage network built on top of IPFS. While IPFS focuses on how data is stored and accessed, Filecoin adds a vital